PROJECT REPORT · SERIES 002

The Classification Engine

Binary classification from first principles applied to diverse domains.

Key Result: **MCC is the most balanced metric for binary classification**

Abstract. This project builds a complete binary classification pipeline from scratch, addressing the challenges of class imbalance and probability calibration. It applies techniques to accurately predict authenticity of manuscripts and the success of silicon chips in testing environments.

GitHub	`https://github.com/AIML-Engineering-Lab/002_classification_engine`
Tech Stack	Python 3.11 · scikit-learn · pandas · Matplotlib
Licence	MIT

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1 Project Overview

This project builds a complete binary classification pipeline from scratch, addressing the challenges of class imbalance and probability calibration. It applies techniques to accurately predict authenticity of manuscripts and the success of silicon chips in testing environments.

1.1 Key Concepts

- Sigmoid Function
- Cross-Entropy Loss
- L1 Regularization
- L2 Regularization
- AUC-ROC

2 Datasets

Dataset	Rows	Features
Ancient Manuscript Authenticity	3000	Ink iron ratio, parchment density, carbon-14 ratio, scribal pressure
Silicon Timing Test Pass/Fail	3000	VDD core voltage, junction temperature, process corner (slow/typic

3 Methodology

3.1 Algorithms

- Sigmoid Function
- Logistic Regression
- SMOTE
- Platt Scaling

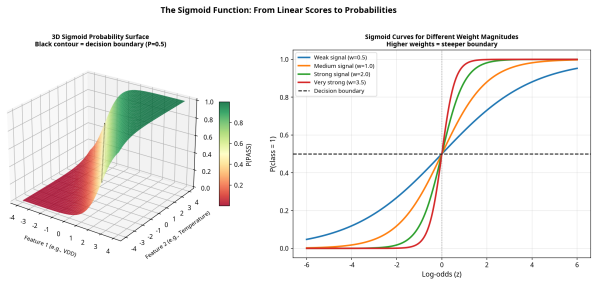
3.2 Processing Pipeline

Raw Data → **EDA** → **Feature Eng.** → **Training** → **Evaluation**

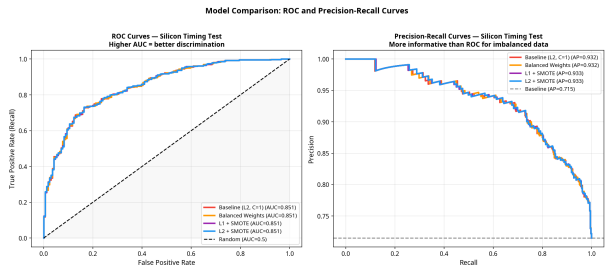
4 Results

Best result: **MCC is the most balanced metric for binary classification**

5 Visualisations

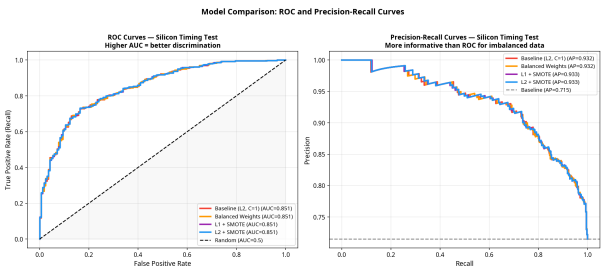


(a) Visualisation 1

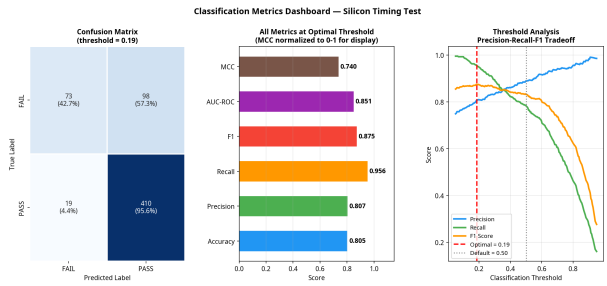


(b) Visualisation 2

Figure 1: Project visualisations



(a) Visualisation 3



(b) Visualisation 4

Figure 2: Additional visualisations

6 Deployment Guide

The project can be deployed using Docker for containerization.

6.1 Quick Start

```
git clone https://github.com/AIML-Engineering-Lab/002_classification_engine
docker compose up --build
```

6.2 Tech Stack

- Python 3.11
- scikit-learn
- pandas
- matplotlib

7 Key Learnings

- Understanding of binary classification frameworks
- Approaches to handle class imbalance
- Techniques for probability calibration

8 Future Work

- Extension to multi-class classification
- Integration of advanced feature engineering methods

9 AIML Engineering Lab Series

This project is part of a progressive series building practical mastery of every major AIML topic from first principles. View the full series at github.com/AIML-Engineering-Lab.